TECHNICAL SPECIFICATION DOCUMENT
MOTEUR T_REX : PROPULSION MIF REBCO

Pulsed Magneto-Inertial System with Passive Trapped-Flux Confinement

Ref. : SPECTRE-MIF-STAP-2026-V3 | Date : May 25, 2026

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Index	Date	Object	Validation
V3.0	25/05/2026	Complete STAP + Closed-Loop Kinetic	NEXUS-6 Eng.

1. EXECUTIVE SUMMARY & TECHNOLOGICAL PARADIGM SHIFT

The T_rex Engine is an advanced, high-energy pulsed magneto-inertial fusion/propulsion (MIF) system designed for deep-space transit. Traditional chemical propulsion architectures (e.g., SpaceX Raptor 3) are strictly bottlenecked by the thermal limits of physical chamber materials and the high molecular mass of exhaust products (H_2O, CO_2). The T_rex Engine bypasses these thermodynamic thresholds by utilizing a 100% passive, ultra-high-field (250 Tesla) magnetic confinement shell generated by High-Temperature Superconducting (HTS) cuprate matrices (REBCO, $YBa_2Cu_3O_{7-x}$).

This architecture completely eliminates the multiphase, highly corrosive flow anomalies caused by lithium hydroxide (LiOH) crystallization encountered in classic Lithium Hydride / Water ($LiH + H_2O$ $LiOH + H_2$) chemical engines. By shifting the physical state of the propulsive medium into a fully ionized Coulomb plasma (Li^{2+}, H^+, O^-) with an ultra-low mean molecular mass (M 3.2 g/mol), the system achieves relativistic exhaust velocities through an asymmetric magnetic nozzle.

2. MECHANICAL ARCHITECTURE & ASYMMETRIC CLOSED-LOOP KINETIC SEQUENCE

Unlike standard MIF concepts, this design does not rely on massive, short-lifespan capacitor banks for direct electrical Z-Pinch ionization. Instead, it utilizes a mechanical Shuttle Piston operating in a closed-loop electromagnetic loop, isolating the fragile HTS components from shock-induced degradation while ensuring a high repetition rate (10 Hz).

2.1 The Shuttle Piston Assembly

The kinetic driver consists of a heavy-duty, reusable hollow capsule engineered from high-yield steel (or prestressed Titanium/Carbon composite). It houses a sealed, cryogenically sustained REBCO HTS permanent magnet operating in a trapped-flux persistent current state. The high-strength steel casing absorbs frontal mechanical impact and acts as a thermal insulator, preserving the superconducting core below its critical temperature ($T < 90$ K) despite external flash plasma temperatures.

2.2 Operational Pulse Sequencing (10 Hz Cycle)

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[Shuttle Rear] ──(Ice Extrusion)──> [Hyperloop Railgun]
      ▲                                    │
      │                                    ▼
[Mag. Deceleration] <── [Field Leak] <── [Frontal Impact]
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- 1. Flash Propellant Extrusion: Rapid automated injector extrudes calibrated pure ice (H_2O solid) onto the shuttle piston faceplate at rear dead center.
- 2. Vac
  Interr

3. MATHEMATICAL FORMALISM & MHD ENGINEERING EQUATIONS

3.1 Passive Magnetic Confinement Pressure

$P_B = B^2 / (2\mu_0)$ [$\mu_0 = 4\pi\times10^{-7}$ H/m]
Application $B = 250$ T → $P_B \approx 24.86\times10^9$ Pa = 24.86 GPa
IP Claim: Passive containment under non-material magnetic pressure 15–30 GPa.

3.2 Plasma Electronic Temperature & Adiabatic Exponent

Following kinetic shock conversion and subsequent adiabatic compression by the 250 T magnetic wall, the core plasma reaches T_e 1×10^6 K. Molecular bonds are utterly broken. The specific heat ratio shifts from its chemical limit (1.28) to the absolute boundary of a fully ionized monoatomic ideal gas: $\gamma = 1.67$.

3.3 Relativistic Jet Exhaust Velocity

$V_e = \sqrt{[(2\gamma)/(\gamma-1) * (R_g * T_e) / M]}$
$R_g = 8.314$ J/(mol·K), $M = 3.2\times10^{-3}$ kg/mol, $T_e = 10^6$ K
$V_e \approx 113\ 800$ m/s = 113.8 km/s
$I_{sp} = V_e / g_0 \approx 11\ 604$ seconds (vacuum)

4. MASS ALLOCATION & STRUCTURAL SPECIFICATIONS

To sustain a persistent 250 T field without catastrophic thermal transition (quench), the magnet assembly is optimized based on REBCO critical current density (J_c) thresholds at 4.2 K.

Subsystem Component	Net Mass	Specific Function	Advanced Material
HTS REBCO Magnet Matrix	1.8–3.5 t	Generation/maintenance of passive [illegible]	High-J_c YBCO/GdBCO tapes + Ag/Cu stabilizer
Lorentz Tension Exoskeleton	2.5–3.0 t	Suppress radial hoop stress under 25[illegible]	Prestressed Carbon T800 + Ti Grade 5 ribbing
Cryostat & Vacuum Jacket	0.8–1.2 t	Cryogenic isolation & thermal monitoring	Invar casing + MLI + He pulse-tubes
DRY MAGNÉTIQUE CORE MASS	5.1–7.7 t	Total Integrated Propulsion Power Pl	Unified Engine Nacelle

5. DEFENSIVE PATENT CLAIMS & PRIOR ART DEFINITIONS

The inventor explicitly claims technological priority and prior art on the following industrial methodologies and configurations as of May 25, 2026:

- 1. Utilization of solid LiH + H_2O interaction not as traditional gaseous chemical combustion, but as a precursor for fully ionized, low-mass propulsive plasma via hyper-velocity mechanical shock.

- 2. Clo
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